

Project Proposal: Toxicity Detection in Social Media Comments Using Fine-Tuned BERT

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Overview

This project fine-tunes a pre-trained BERT model on the Jigsaw Toxic Comment Classification dataset to detect toxic content in social media comments. The goal is to build a reliable multi-label classifier that identifies toxicity categories such as *toxic*, *severe toxic*, *obscene*, *threat*, *insult*, and *identity hate*.

Dataset

The Jigsaw Toxic Comment Classification dataset (Kaggle) contains ~160K Wikipedia talk page comments, each labeled across six toxicity categories. Preprocessing will include text cleaning, tokenization using `BertTokenizer`, and handling class imbalance via weighted loss.

Methodology

- Model: `bert-base-uncased` with a custom multi-label classification head
- Fine-tuning: Freeze lower BERT layers; train upper layers and classifier head
- Loss Function: Binary Cross-Entropy with class weights to address label imbalance
- Optimizer: AdamW with linear learning rate scheduler
- Framework: PyTorch + HuggingFace Transformers

Evaluation Metrics

- ROC-AUC (per label and macro-averaged)
- F1-Score (micro and macro)
- Precision-Recall curves for each toxicity class